

OPTIMIZING MARITIME LOGISTICS AND TRADE OPERATIONS



CHALLENGE 7.1: Coastal Vessel Optimization- Minimizing Bulk Cargo Transportation Cost

Background & Context

Optimizing coastal tanker allocation is critical for cost-effective bulk cargo movement in HPCL's international and coastal trade operations. The objective is to develop an optimal transportation model for coastal shipping of bulk cargo using a fleet of 9 coastal tankers, 6 loading ports, and 11 unloading ports. The goal is to minimize the total transportation cost while satisfying demand at all unloading ports under a set of operational and logistical constraints.

Each trip undertaken by a coastal tanker incurs a transportation cost, calculated as the product of the charter hire rate per day (which varies by tanker) and the total number of days consumed in the trip. A trip is defined as the entire journey from a single loading port to one or two discharge ports. The trip duration is dependent on the specific loading and unloading port pair and the tanker used, and this data is available. Additionally, inter-port sailing times between unloading ports are also available, which may be relevant for tanker routing or repositioning.

The transportation model must adhere to the following constraints:

1. Single-port full loading: Each tanker must load its full capacity from only one loading port. Multiport loading is not allowed.
2. Maximum two-port discharge: A tanker may unload at a maximum of two unloading ports per trip.
3. Unlimited supply: Loading ports have unlimited cargo availability, so supply is not a limiting factor.
4. Demand satisfaction: The demand at all unloading ports must be fully met.

Challenge Statement

Develop a transportation model that assigns tankers to trips, determines loading and discharge ports, calculate the number of trips and voyage order, and compute minimum achievable transportation cost across the system. The solution should minimize total transportation cost while ensuring all unloading port demands are met and all operational constraints are respected.

Expected Outcomes

- **Tanker Assignment:** Optimal allocation of tankers to trips, specifying loading port, discharge ports, and trip sequence.
- **Trip Planning:** Number of trips required per tanker to satisfy all demand.
- **Cargo Allocation:** Volume of cargo assigned to each unloading port per trip.

Cost Minimization: Minimum achievable transportation cost for the system while meeting all operational requirements.

Input Tables & Output Format

Loading Ports (L)	Unloading Ports (U)	Tankers (T)	Tanker Capacities (MT)	Charter Hire per Day (Rs Cr/day)	Demand at Unloading Ports (MT/month)
L1	U1	T1	50000	0.63	40000
L2	U2	T2	50000	0.49	135000
L3	U3	T3	50000	0.51	5000
L4	U4	T4	50000	0.51	20000
L5	U5	T5	50000	0.53	20000
L6	U6	T6	50000	0.57	20000
	U7	T7	50000	0.65	110000
	U8	T8	25000	0.39	30000
	U9	T9	25000	0.38	20000
	U10				20000
	U11				20000

Trip time table [Load port to Discharge Port (trip time in days)]

Trip Times (L to U)	U1	U2	U3	U4	U5	U6	U7	U8	U9	U10	U11
L1	0.4	0.7	0.4	0.4	0.4	0.6	0.5	0.4	0.3	0.5	0.7
L2	0.4	0.6	0.5	0.4	0.4	0.5	0.5	0.5	0.3	0.5	0.6
L3	0.4	0.6	0.5	0.4	0.4	0.5	0.5	0.5	0.3	0.6	0.6
L4	0.4	0.6	0.4	0.3	0.3	0.5	0.5	0.4	0.3	0.5	0.6
L5	0.4	0.6	0.4	0.3	0.3	0.5	0.5	0.4	0.3	0.5	0.5
L6	0.58	0.73	0.64	0.56	0.56	0.65	0.67	0.64	0.50	0.70	0.73

Trip times b/w discharge ports (trip time in days)

Trip Times	U1	U2	U3	U4	U5	U6	U7	U8	U9	U10	U11
U1	0.00	0.35	0.19	0.16	0.16	0.25	0.23	0.19	0.09	0.19	0.33
U2	0.36	0.00	0.38	0.37	0.37	0.15	0.34	0.38	0.29	0.40	0.19
U3	0.16	0.33	0.00	0.16	0.16	0.25	0.21	0.17	0.09	0.19	0.32
U4	0.18	0.37	0.21	0.00	0.12	0.29	0.25	0.21	0.05	0.21	0.35
U5	0.18	0.37	0.21	0.12	0.00	0.29	0.25	0.21	0.05	0.21	0.35
U6	0.29	0.17	0.32	0.31	0.31	0.00	0.22	0.32	0.24	0.34	0.18
U7	0.19	0.29	0.21	0.20	0.20	0.15	0.00	0.21	0.12	0.21	0.28
U8	0.16	0.33	0.17	0.16	0.16	0.25	0.21	0.00	0.09	0.19	0.32
U9	0.17	0.37	0.21	0.12	0.12	0.29	0.25	0.21	0.00	0.21	0.35
U10	0.14	0.34	0.18	0.15	0.15	0.27	0.20	0.18	0.07	0.00	0.33
U11	0.33	0.17	0.34	0.33	0.33	0.13	0.30	0.34	0.25	0.36	0.00

Output format (example)

Source	Destination	Tanker	Volume (MT)	Trip Cost (Rs Cr)